Model Validation Report

Pima Indians Diabetes Dataset

AnalystLab Africa Machine Learning Internship — Week 6

1. Dataset Description

- The dataset is the Pima Indians Diabetes dataset, sourced from the National Institute of Diabetes and Digestive and Kidney Diseases.
- The data represents female patients of Pima Indian heritage who are at least 21 years old.
- The total number of records is 768, containing 8 predictor variables and 1 binary target variable.
- The class distribution consists of 500 negative cases and 268 positive cases, resulting in an approximate 34.9% positive rate.
- There are no detected missing values or duplicate rows in the dataset.
- The data was split into 70% training data (537 samples) and 30% test data (231 samples) using a random state of 43.
- Feature scaling was not applied, as it is not required for Decision Tree models.

2. Baseline Model Performance

- The baseline model was a Decision Tree Classifier using default scikit-learn parameters.
- The default configuration resulted in a fully grown tree that was prone to overfitting.
- **Accuracy:** 0.65 (65% of predictions were correct)
- **Precision:** 0.49 (49% of positive predictions were actually positive)
- **Recall:** 0.50 (50% of actual positives were correctly identified)
- **F1-Score:** 0.50
- **ROC-AUC:** ~0.60

3. Hyperparameters Tested

- Hyperparameter tuning utilized both Grid Search and Randomized Search strategies with 5-fold cross-validation.
- `max_depth` values tested were 2, 5, 7, 10, 15, 20, 25, 50, and 100 for Grid Search, and a range of 2–137 for Randomized Search.
- `min_samples_split` values tested were 2, 3, 4, and 5.
- `min_samples_leaf` values tested were 1, 2, 3, 4, and 5.
- `max_features` values tested were 1, 2, 3, 4, and 5.
- The `class_weight` parameter was set to "balanced".

- Grid Search exhaustively explored 900 combinations, while Randomized Search sampled 20 combinations.

4. Best Parameters Identified

- Both Grid Search and Randomized Search converged on the exact same optimal parameter configuration.
- **Best max_depth:** 100
- **Best max_features:** 5
- **Best min_samples_leaf:** 4
- **Best min_samples_split:** 5
- **Best class_weight:** "balanced"
- Randomized Search found these parameters in ~447 milliseconds, while Grid Search took ~19 seconds.

5. Cross-Validation Results

- Internal hyperparameter tuning split the 537 training samples into 5 folds of approximately 107 samples each.
- Each parameter combination was trained on 4 folds and validated on the 5th fold, repeating this process 5 times.
- Standalone cross-validation included a default 5-fold `cross_val_score` for the baseline model.
- Standalone K-Fold Cross-Validation utilized 6 splits for stratified evaluation to ensure each fold was tested exactly once.
- Cross-validation is crucial for this dataset due to its small size and the ~65/35 class imbalance.

6. Final Model Performance

Metric	Grid Search Tuned Model	Randomized Search Tuned Model
Accuracy	0.74	0.71
Precision	0.66	0.60
Recall	0.53	0.49
F1-Score	0.59	0.54
ROC-AUC	~0.70	~0.66

Note: Grid Search yielded slightly better final test performance despite identifying the same best parameters as Randomized Search.

7. Key Findings & Recommendations

- **Overfitting Mitigated:** Hyperparameter tuning successfully mitigated overfitting and delivered substantial performance gains, boosting Grid Search accuracy to 74% and F1-Score to 0.59.
- **Efficiency of Randomized Search:** Randomized Search proved highly efficient, matching Grid Search optimal parameters while exploring only 2.2% of the parameter space.
- **Handling Class Imbalance:** Setting `class_weight='balanced'` was crucial for handling class imbalances, and Precision saw a much larger improvement (+35%) than Recall (+6%).
- **Future Recommendation — Imputation:** Future iterations should address feature preprocessing to handle physiologically impossible zero values through imputation.
- **Future Recommendation — Ensembles:** Future modeling should transition to ensemble methods like Random Forest or Gradient Boosting to further reduce variance.
- **Future Recommendation — Optimize for Recall:** Future model tuning should optimize specifically for Recall to minimize dangerous false negative predictions in the medical context.
- **Future Recommendation — Validation Strategy:** Future evaluations should leverage Stratified K-Fold cross-validation to maintain class distribution across folds.
- **Future Recommendation — Tuning Limits:** Future tuning efforts should prioritize Randomized Search utilizing wider ranges, allowing for efficient exploration of larger parameter spaces without computational lag.